

SMART LEAVE APPROVAL SYSTEM

Architecture, Design, User Guide & AI Change-Loop Evidence

Project Documentation Package

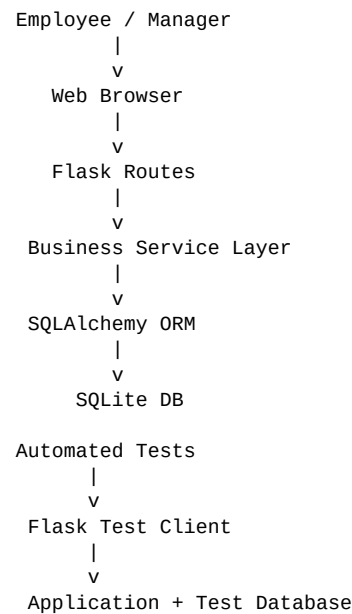
This document combines the system architecture, application design, user guide, and AI-assisted change-loop evidence for the Smart Leave Approval System assessment.

1. System Architecture

The application follows a simple layered web architecture. The browser provides the user interface, Flask handles HTTP requests and routing, the service layer applies leave business rules, SQLAlchemy provides database access, and SQLite stores leave requests.

Layer	Responsibility	Technology
Presentation	Forms, dashboard, status messages	HTML, Bootstrap
Application	Routes and request handling	Flask
Business	Leave validation and approval rules	Python service layer
Data Access	ORM models and database operations	SQLAlchemy
Database	Persistent leave request storage	SQLite
Testing	Automated functional and validation tests	pytest

2. Architecture Flow



3. Application Design

The design intentionally focuses on one complete workflow rather than many shallow features. The employee creates a leave request. The server validates the request before saving it. The manager dashboard displays pending requests and provides approval or rejection actions.

Component	Design
Employee Form	Collects employee name, start date, end date and reason.
Validation	Applies all leave rules on the server.
Leave Model	Stores request details, status and creation time.
Manager Dashboard	Lists pending requests and decision actions.
Approval Flow	PENDING → APPROVED or PENDING → REJECTED.

Component	Design
Cancellation	PENDING → CANCELLED; finalized requests cannot be cancelled.

4. Database Design

Main entity: LeaveRequest

Field	Purpose
id	Unique leave request identifier
employee_name	Name of employee requesting leave
start_date	First day of leave
end_date	Last day of leave
reason	Reason for leave
status	PENDING, APPROVED, REJECTED or CANCELLED
created_at	Time at which request was created

5. User Guide — Employee

- Open the application home page.
- Select the option to apply for leave.
- Enter the employee name.
- Select a valid start date and end date.
- Enter a reason when leave is longer than five days.
- Submit the leave request.
- Review the request status after submission.
- If the request is still PENDING, use the cancellation option when cancellation is required.

6. User Guide — Manager

- Open the Manager Dashboard.
- Review pending leave requests.
- Check employee name, dates, duration and reason.
- Select Approve to approve a valid pending request.
- Select Reject to reject a pending request.
- After a decision, the request becomes finalized and cannot be processed again.

7. Validation Messages

Condition	Expected behavior
Missing employee name	Reject submission
Past start date	Reject submission
End date before start date	Reject submission
More than 10 days	Reject submission
More than 5 days without reason	Reject submission
Overlapping leave	Reject submission
Valid request	Create PENDING request

8. Installation and Running Guide

```
# Create virtual environment
python -m venv venv
```

```
# Activate on Windows
venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Start application
python run.py

# Run automated tests
pytest -q
```

9. AI Change-Loop Evidence

The change-loop feature used for this project is leave cancellation. The AI is instructed to implement the feature, execute the existing tests, identify failures, correct the implementation, add regression tests, and execute the complete suite again.

Initial feature prompt

Add cancellation support to the Smart Leave Approval System.

Allow an employee to cancel a leave request only when its status is PENDING.

Requirements:

- Add CANCELLED status.
- Allow PENDING -> CANCELLED.
- Do not allow APPROVED or REJECTED -> CANCELLED.
- Do not allow CANCELLED -> APPROVED or REJECTED.
- Update the UI.
- Preserve all existing business rules.
- Run the existing pytest suite.
- Add regression tests.
- Fix failures without weakening tests.

10. Failure and Correction Log

Attempt	Action	Evidence to record
1	AI implements cancellation	Prompt, changed files and diff
2	Existing tests are executed	Actual pytest output
3	AI diagnoses failures	Failure explanation
4	AI corrects implementation	Corrective diff
5	Regression tests added	New test cases
6	Full suite executed	Actual final pytest output

11. Evidence Checklist

- Architecture and application design
- Employee application screenshot
- Manager dashboard screenshot
- Validation-error screenshot
- Approval/rejection screenshot
- AI feature prompt
- AI code changes or git diff
- First test run after the feature
- Failure/red test output, if produced
- Correction prompt and response
- New regression tests
- Final green test output

12. Conclusion

The documentation demonstrates the complete engineering workflow for the Smart Leave Approval System: architecture and design, user operation, automated testing, AI-assisted feature development, failure analysis, correction, and final verification. Actual screenshots and terminal output should be inserted wherever indicated before submitting the assessment.